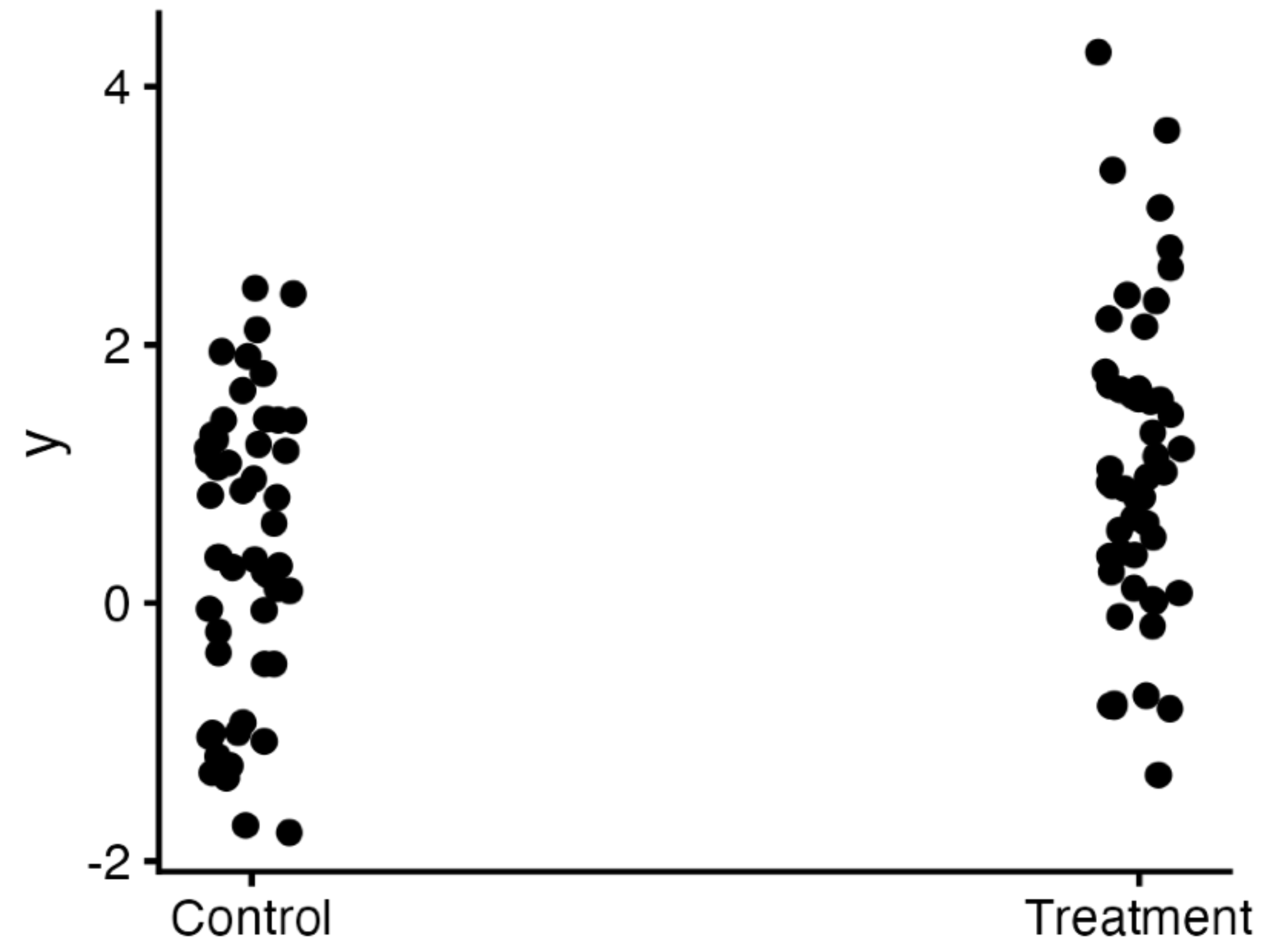


What about categorical predictors?

Linear regression is flexible

- Our questions are frequently based on categories:
 - Is a treatment effective in improving outcomes?
 - Are two geographical regions different in some aspect?
 - Does the diet of a group of species affect their size?



Binary predictor, same model

Control-treatment, two categories...

$$y_i \sim N(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta x_i$$

- x_i : 0 for control, 1 for treatment
- α : the intercept is the **mean of the Control** group
- β : the slope is the **difference in the means across the groups**

```
> precis(fit, prob = 0.95)
      mean    sd 2.5% 97.5%
a      0.52 0.16 0.20  0.82
b      1.37 0.22 0.96  1.79
sigma 1.08 0.08 0.94  1.23
```

